

Section 2 Logical Thinking & Flowcharts

Flowcharts

- ✓ Diagrammatic representation
- ✓ Break down any process into smaller steps

Components of Flowchart

or
Building Blocks of Flowchart

Start

End

Starters / Terminators

Read N

input/output

Print N

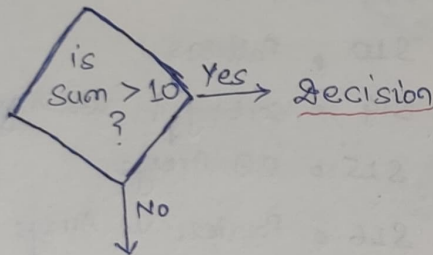
Print "Hi"

Let Sum = 0

Process

= Vs =
↑
assignment

= Vs =
↑
Equality



Arrows are use to connect components in Flowchart

Flowchart - Simple Interest Calculator

Read Principal, Rate and Time and print Simple Interest.

Sample Input

P = 100 ₹

R = 5%

T = 2 years

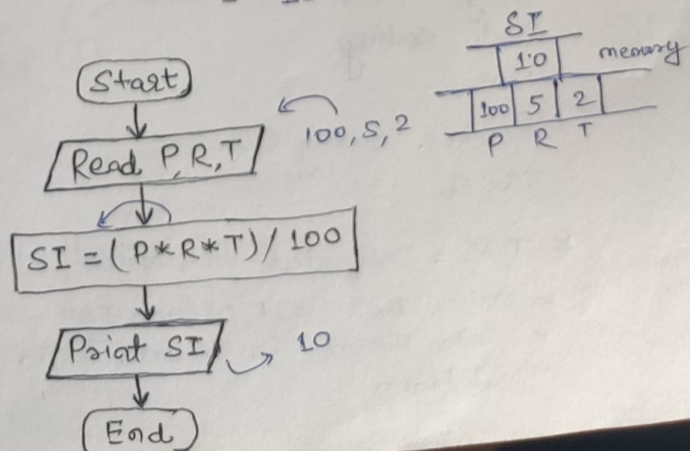
$$SI = \frac{P \times R \times T}{100}$$

$$= \frac{100 \times 5 \times 2}{100}$$

$$= 10$$

Sample output

SI = 10



Flowchart - Largest Number

Given 3 numbers, find the largest number.

Sample Input

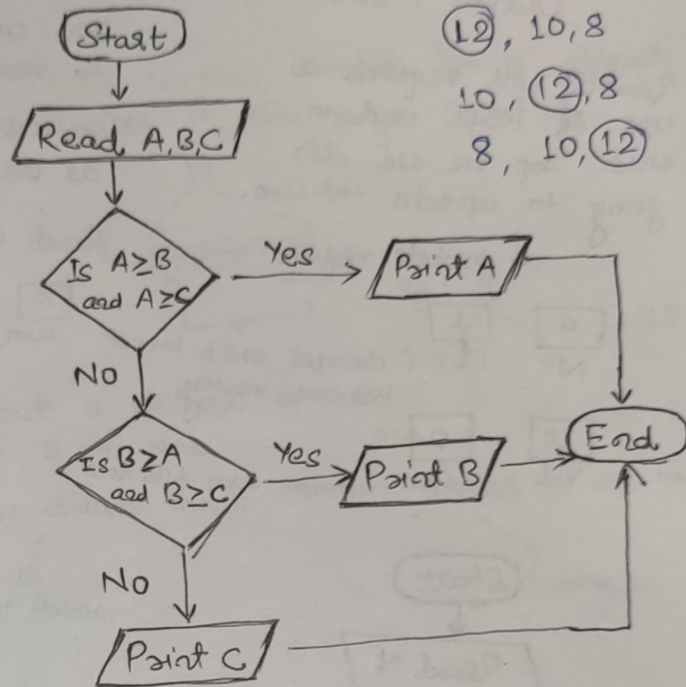
A = 10

B = 30

C = 20

Sample Output

30



Flowchart - ~~Number~~ Sum of First N Number

Find the sum of numbers from 1 to N.

Example

N = 4

Output

10

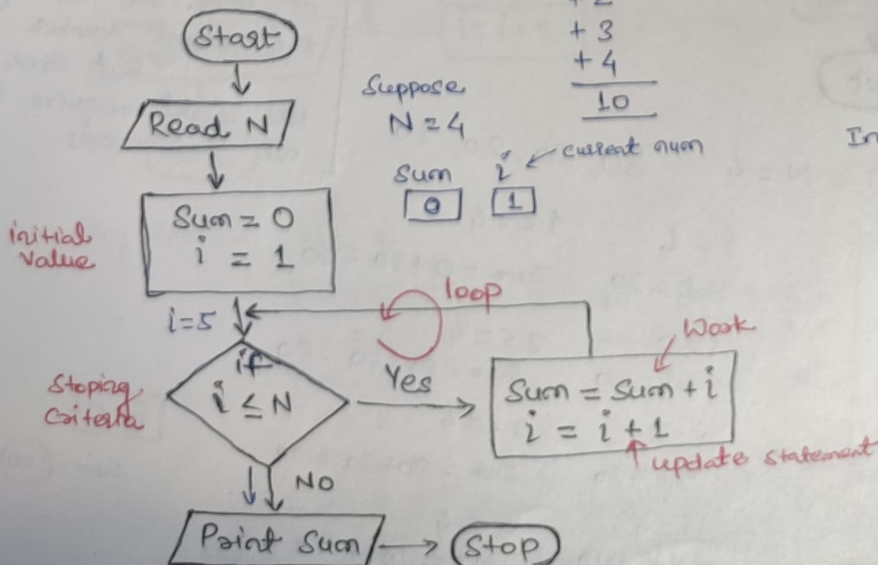
$$1 + 2 + 3 + 4 = 10$$

Repeating work

Sum = 0

$$\begin{array}{r}
 +1 \\
 +2 \\
 +3 \\
 +4 \\
 \hline
 10
 \end{array}$$

Loop



Initially

Sum	1
-----	---

1 ≤ 4	Sum	2
2 ≤ 4	Sum	3
3 ≤ 4	Sum	4
4 ≤ 4	Sum	5

Flowchart - Sum of Multiple Inputs

Take input N followed by N more numbers, find the sum of those numbers.

eg: $N = 4$

loop $\boxed{10, 20, 30, 40}$
 $i=1 \quad i=2 \quad i=3 \quad i=4$
 Output: 100

Reading will require a use of loop and in this loop we are also going to update the sum.

4 buckets require

$\boxed{4}$ $\boxed{1}$
 N $\downarrow$ i denotes which bucket we are reading
 $\boxed{10}$ $\boxed{0+10}$
 Val Sum denotes sum till now

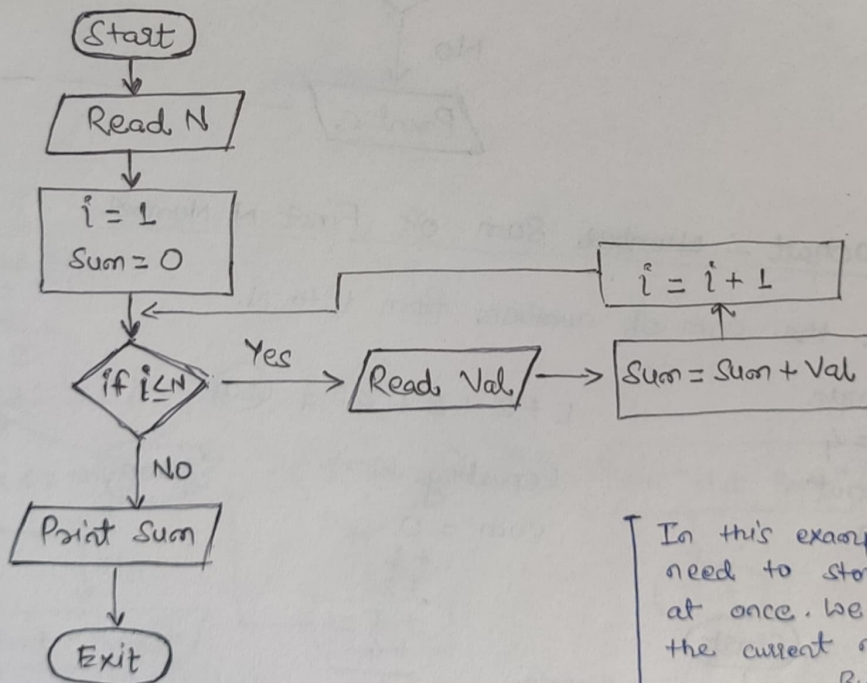
N could be anything if $N = 100$

You can't create N number of buckets to read such number.

So, we have to read & process the sum as we read through input.

Read and Update Sum

$\boxed{0}$ $0+10 = 10+20 = 30+30 = 60+40 = 100$
 Sum



In this example, we do not need to store all the numbers at once. We are only storing the current number.

But some problems we might require to store all the numbers

Dry Run: $N = 4$ 30, 50, 20, 10

$i = 1$ $1 \leq 4$
 $Val = 30$ $Sum = 0 + 30 = 30$
 $i = 2$ $2 \leq 4$
 $Val = 50$ $Sum = 30 + 50 = 80$
 $i = 3$ $3 \leq 4$
 $Val = 20$ $Sum = 80 + 20 = 100$
 $i = 4$ $4 \leq 4$
 $Val = 10$ $Sum = 100 + 10 = 110$

$\boxed{i = 5}$
 $Sum = 100$

Flowchart - Prime Number

Given a Number, check if it is Prime or Not!

Input

11

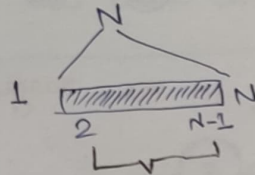
Output

Yes

Prime Number is a number that can only be divisible by 1 and itself.

eg: 2, 3, 5, 7, 11, 13

Non-Prime Number, number having atleast one divisor in the range 2 to $N-1$



eg: 9
2 3 ... 8

$$9 \% 3 = 0$$

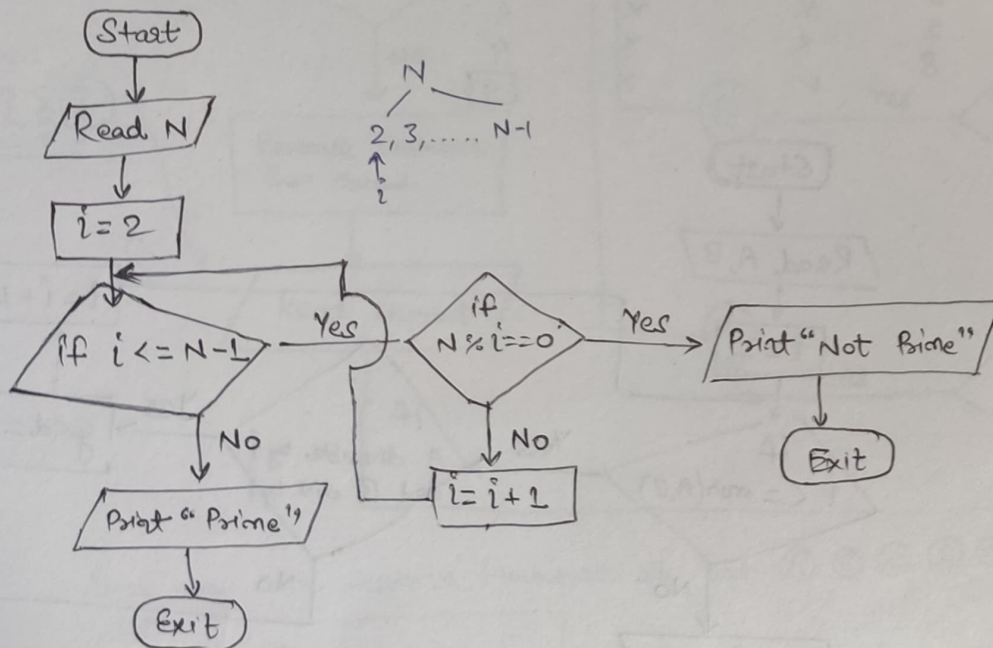
Logic :- Find out a divisor in the range 2 to $N-1$

If Yes, divisor found

Not Prime

If divisor is not found

Prime.



Dry Run :-

$N = 7$			
i	N		
$2 <= 6$	$7 \% 2$	No	$i = i + 1 = 3$
$3 <= 6$	$7 \% 3$	No	$3 + 1 = 4$
$4 <= 6$	$7 \% 4$	No	$4 + 1 = 5$
$5 <= 6$	$7 \% 5$	No	$5 + 1 = 6$
$6 <= 6$	$7 \% 6$	No	$6 + 1 = 7$
$7 <= 6$			

Hence 7 is Prime Number

$N = 9$

i	N		
$2 \leq 9$	$9 \% 2$	No	$i = 2 + 1 = 3$
$3 \leq 9$	$9 \% 3$	= 0	Yes

Hence, 9 is not prime

Flowchart - GCD / HCF

Find GCD (Greatest Common Divisor) of two numbers.

$$a = 8$$

$$b = 12$$

$$\text{GCD} \rightarrow 4$$

$$a = 8$$

$$b = 24$$

$$\text{GCD} \rightarrow 8$$

$$a = 9$$

$$b = 11$$

$$\text{GCD} \rightarrow 1$$

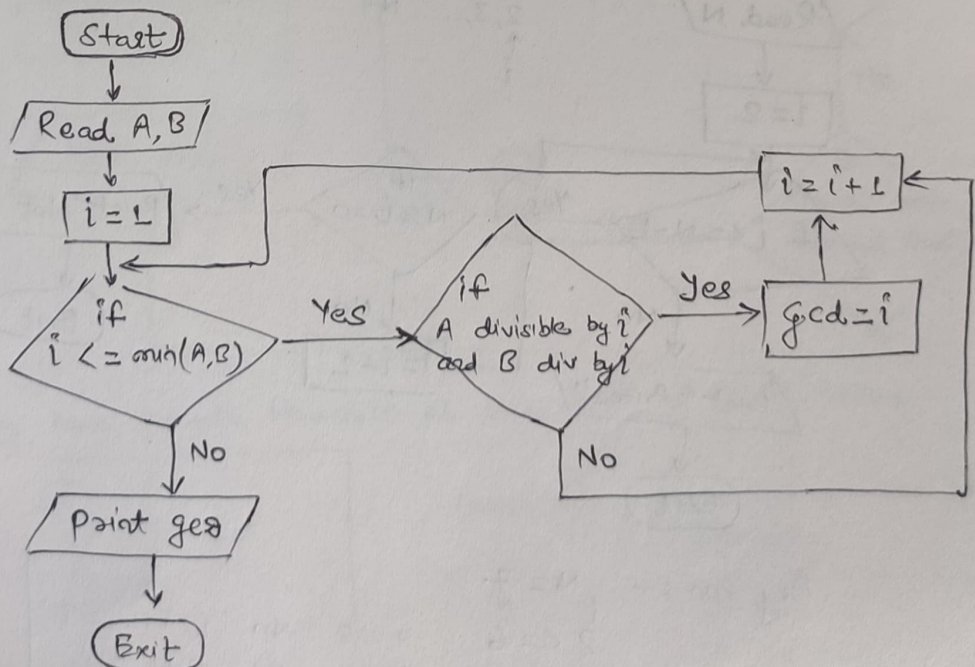
GCD(a, b)

$$a, b \quad \text{gcd}(a, b) \leq \min(a, b)$$

$$\begin{cases} a = 8 \\ b = 12 \end{cases} \quad \text{X } 10$$

GCD could lie in the range 1 to $\min(a, b)$
└──────────┘
loop

	a = 8	b = 20	a = 8	b = 20	gcd
Range					
1	✓	✓	✓	✓	1
2	✓	✓	✓	✓	2
3	X	X	X	X	3
4	✓	✓	✓	✓	4
5	X	✓	X	✓	5
6	X	X	X	X	6
7	X	X	X	X	7
8	✓	X	✓	X	8



dry Run :

	a	b
1	✓	✓
2	✓	✓
3	✓	X
4	X	✓

Range 1 to 4

is 1 divisible by 6

$$\text{gcd} = 1$$

$$\text{gcd} = 2$$

$$\text{gcd} = 2$$

$$\text{gcd} = 2$$

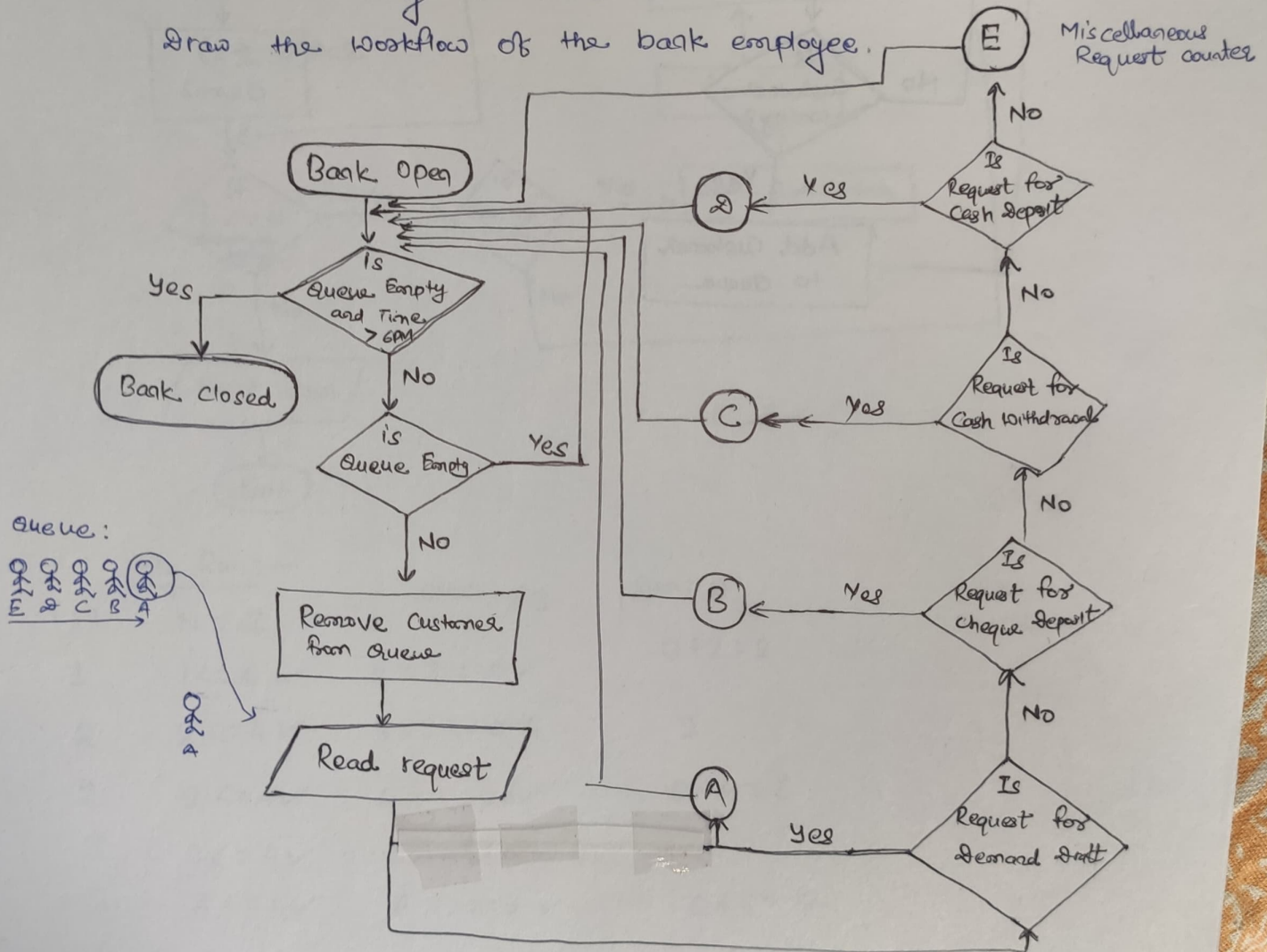
Flowchart - Bank Employee

A Bank is open till 6PM. and the banker needs to process requests from customers.

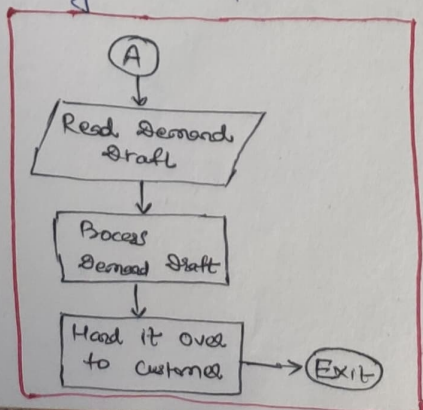
Requests can be one of the four types:-

- Cash Deposit
- Cash withdrawal
- cheque deposit
- Making A Demand Draft

Draw the workflow of the bank employee.



You may have separate flowchart of each A B C D E Actions

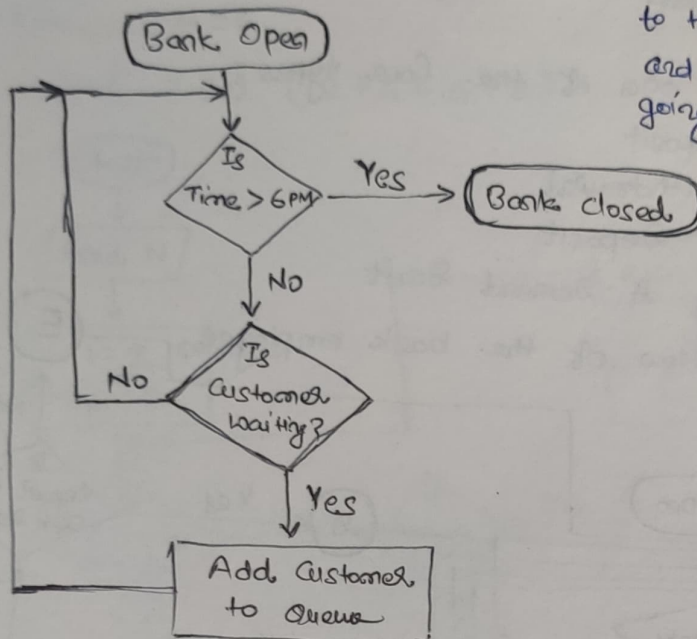


Connectors → They combine two different flowcharts.

Flowchart - Bank Guard

Flowchart of Bank Security Guard,

So, the bank security guard is going to wait for the customers outside the bank, if they're coming before 6:00PM, He will add the customer to the queue. otherwise keep on waiting and if time reaches 6PM. He is going to close the bank.

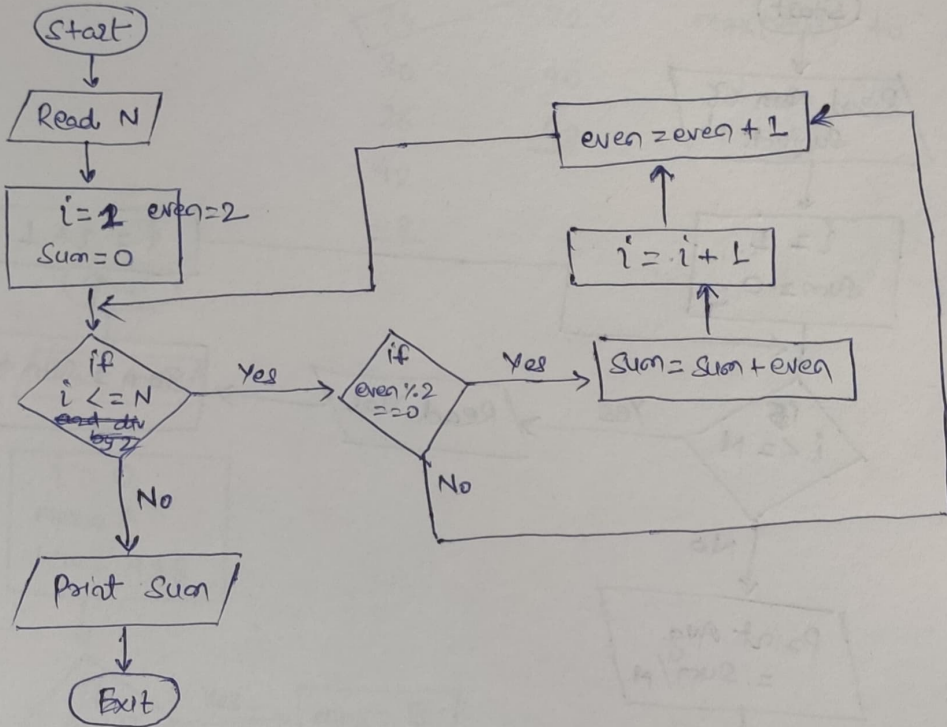


Flowchart Assignment

1. Draw a flowchart to find sum of first N even numbers starting from 2.

Example $N=4$

Output = 20 ($2+4+6+8$)



Dry Run:-

$i=1$	$N=4$	$even=2$	$Sum=0$
1	$1 \leq 4 \checkmark$	$2 \% 2 == 0 \checkmark$	$0+2=2$
2	$2 \leq 4 \checkmark$	$3 \% 2 == 0 \times$	2
2	$2 \leq 4 \checkmark$	$4 \% 2 == 0 \checkmark$	$2+4=6$
3	$3 \leq 4 \checkmark$	$5 \% 2 == 0 \times$	6
3	$3 \leq 4 \checkmark$	$6 \% 2 == 0 \checkmark$	$6+6=12$
4	$4 \leq 4 \checkmark$	$7 \% 2 == 0 \times$	12
4	$4 \leq 4 \checkmark$	$8 \% 2 == 0 \checkmark$	$12+8=20$
5	$5 \leq 4 \times$		

↓

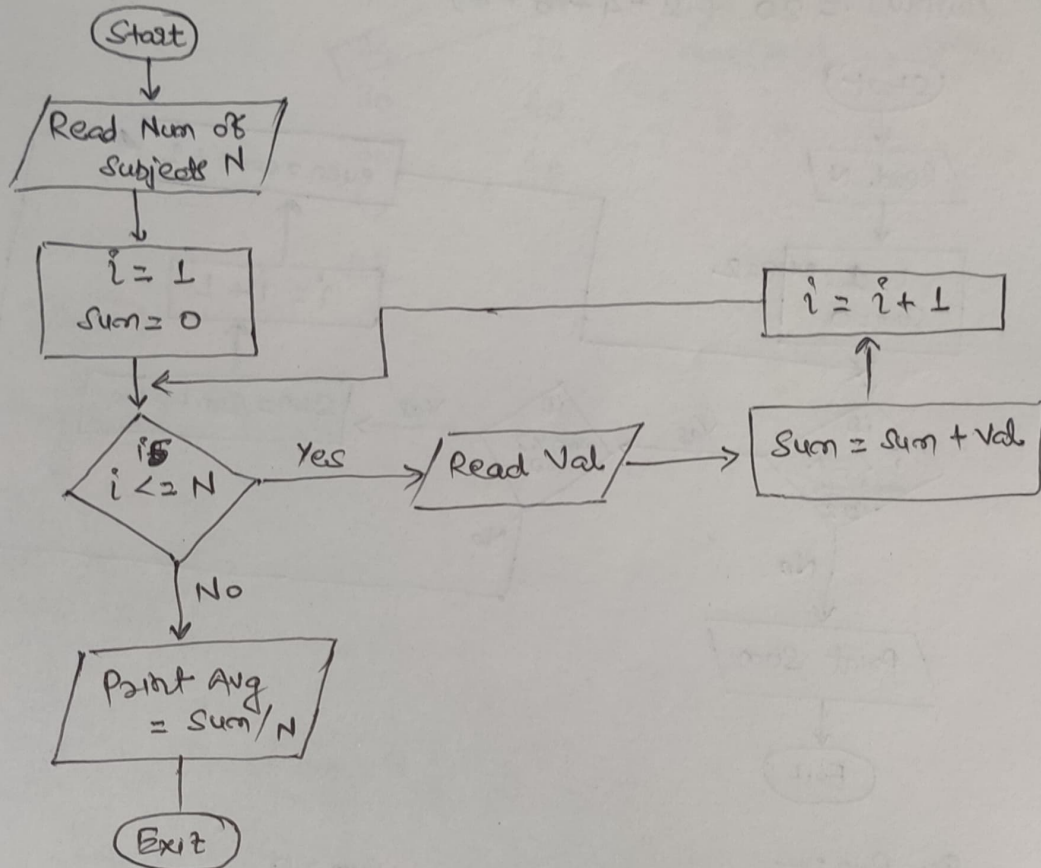
20

2. Draw a flowchart to input marks in 5 subjects and find their average.

Input: 5

100 80 90 70 90

output: 86 $Avg = \frac{100 + 80 + 90 + 70 + 90}{5} = 86$



Try Run

N = 5

100 80 90 70 90

i	i ≤ 5	Val	Sum
1	1 ≤ 5 ✓	100	0 + 100 = 100
2	2 ≤ 5 ✓	80	100 + 80 = 180
3	3 ≤ 5 ✓	90	180 + 90 = 270
4	4 ≤ 5 ✓	70	270 + 70 = 340
5	5 ≤ 5 ✓	90	340 + 90 = 430
6	6 ≤ 5 X		

$$Avg = \frac{430}{5} = 86$$

3. Draw a flowchart to find LCM of two numbers A and B.

You may use the formula $LCM \times HCF = \text{Product}(A, B)$

Input: 6 8

output: 24

A = 8

B = 8

Lowest Common Multiplication

12

16

18

24

24

32

30

40

36

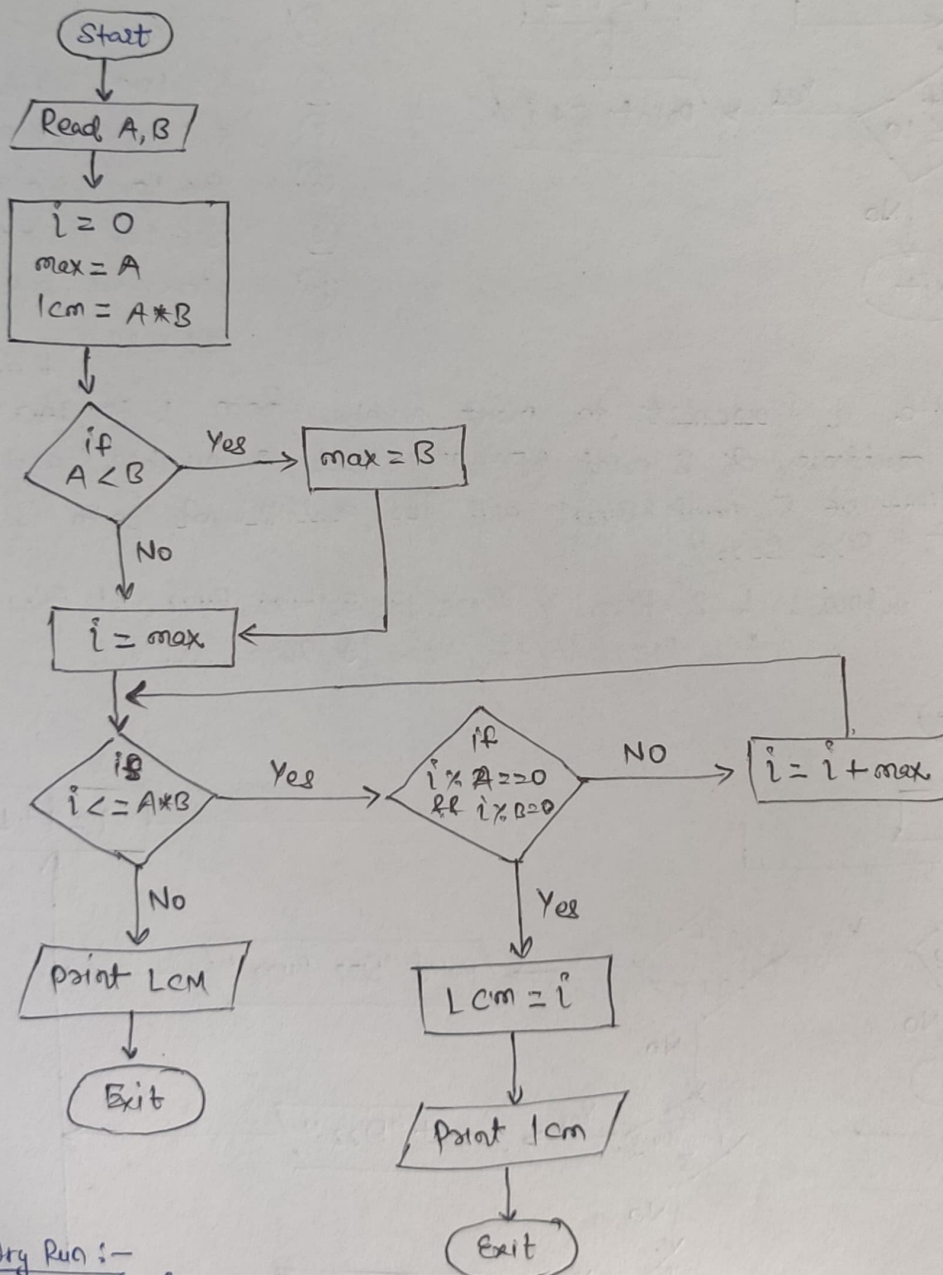
48

42

48

LCM of (A,B) ranges betⁿ $\max(A,B)$ to $A \times B$.

8 to 48



dry Run :-

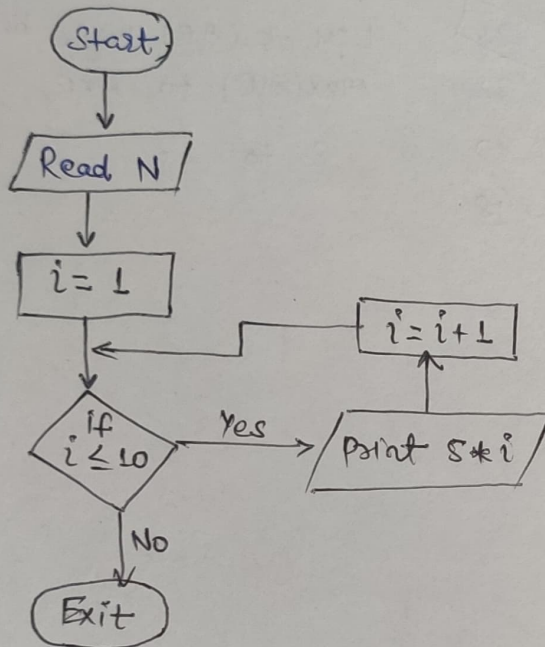
i	A	B	i <= A * B	i % A == 0 & i % B == 0	LCM
0	6	8	0 <= 48 ✓	0 % 6 == 0 & 0 % 8 == 0 ✓	0
8	6	8	8 <= 48 ✓	8 % 6 != 0 ✗	
16	6	8	16 <= 48 ✓	16 % 6 != 0 ✗	
24	6	8	24 <= 48 ✓	24 % 6 == 0 & 24 % 8 == 0 ✓	24

LCM = 6 * 8 = 48

4. Draw a Flowchart to print table of a given Number upto 10.

Example $N = 5$

Output : 5, 10, 15, 20, 25, 30, 35, 40, 45, 50.



Dry Run :-

	$N = 5$	
$i = ①$	$1 \leq 10 \checkmark$	$5 * 1 = 5$
$i = ②$	$2 \leq 10 \checkmark$	$5 * 2 = 10$
③	$3 \leq 10 \checkmark$	$5 * 3 = 15$
④	$4 \leq 10 \checkmark$	$5 * 4 = 20$
⑤	$5 \leq 10 \checkmark$	$5 * 5 = 25$
⑥	$6 \leq 10 \checkmark$	$5 * 6 = 30$
⑦	$7 \leq 10 \checkmark$	$5 * 7 = 35$
⑧	$8 \leq 10 \checkmark$	$5 * 8 = 40$
⑨	$9 \leq 10 \checkmark$	$5 * 9 = 45$
⑩	$10 \leq 10 \checkmark$	$5 * 10 = 50$
⑪	$11 \leq 10 \times$	exit

5. Write a Flowchart to print numbers from 1 to 100. and for multiple of 3 print "Fizz" instead of number and for multiple of 5 print "Buzz" and for multiple of both 3 & 5 print "Fizz Buzz".

Sample output : 1, 2, Fizz, 4, Buzz, 7, 8, Fizz, Buzz, 11, Fizz, 13, 14, Fizz Buzz, 16, 17, Fizz, 19, Buzz, ...

